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RECEIVED 29 January 2026

REVISED 01 March 2026

ACCEPTED 02 March 2026

PUBLISHED 29 April 2026

### CITATION

Fan P, Chen Z and Wang Y (2026)  
Research on photovoltaic power  
generation based on multi-dimensional  
indicators and models.  
*Front. Environ. Sci.* 14:1799258.  
doi: 10.3389/fenvs.2026.1799258

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# Research on photovoltaic power generation based on multi-dimensional indicators and models

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**Introduction:** Photovoltaic (PV) power generation is vital for sustainable energy and carbon reduction, yet existing studies often focus on single aspects, lacking integrated planning support.

**Methods:** This study develops a framework combining power forecasting, optimization, and carbon assessment using a multidimensional indicator system, PCA, t-SNE, and PSO.

**Results:** A 1% increase in PV generation could reduce China's power sector carbon emissions by 2.05% by 2035; the model achieved an  $R^2=0.9975$ .

**Discussion:** The framework supports regional energy planning, though future models should incorporate policy shifts and market dynamics.

### KEYWORDS

carbon emission prediction, multidimensional indicators, particle swarm optimization, photovoltaic power generation, principal component analysis, sustainable development

## 1 Introduction

As one of the world's largest energy consumers, China's power supply system is crucial to the stability of its domestic economy and society. In recent years, with the rapid economic development and the advancement of industrialization and urbanization, the demand for electricity has continued to grow, and higher requirements have been placed on power generation supply (Nie et al., 2024). At present, the power generation structure in China is diversified. Traditional energy generation, especially coal power generation, still dominates because of its abundant reserves and low price, but it brings serious environmental problems and faces environmental protection pressure (Zheng et al., 2025). Meanwhile, the Chinese government is actively developing renewable energy sources such as solar and wind power, among which photovoltaic power generation has made significant progress. China has become one of the largest photovoltaic markets in the world, with power stations all over the country, especially in the western region. The growth in photovoltaic power helps reduce reliance on traditional energy sources and supports carbon reduction targets. However, photovoltaic power faces challenges such as being greatly affected by the weather, stability and reliability need to be addressed, and high construction and operation costs require technological innovation and policy support to reduce costs and address environmental issues (Rahman et al., 2025). In addition, China's power supply is unevenly distributed across regions, with high demand but scarce resources in the eastern coastal areas and abundant resources but low demand in the western regions. Therefore, strengthening the construction of power transmission and distribution networks and achieving cross-regional interconnection are important ways to solve the contradiction between regional power supply and demand. In conclusion, China's power supply system is facing an important

period of transition from traditional energy to clean energy. Although coal power generation still dominates, the rapid development of renewable energy such as photovoltaic power generation brings new hope for future power supply. The government needs to continue to increase policy support, promote technological innovation and market development to achieve sustainable development of power supply and green transformation of the economy and society.

## 2 Literature review

Photovoltaic (PV) power generation, as an important clean energy source, has gained widespread attention globally in recent years. Despite continuous technological advancements, PV power generation still faces several challenges, such as poor stability, high costs, and environmental impacts. Existing research has explored various aspects of PV power generation, including efficiency, economic viability, and carbon emissions, but numerous challenges remain to be addressed, particularly in improving power generation efficiency, reducing costs, and mitigating carbon emissions. In addition, regional differences in electricity demand and resource distribution, as well as the policy and market environment, have also had a significant impact on the development of PV power generation. In the following sections, we will review the relevant literature, analyze the limitations of existing research, and discuss potential future research directions.

### 2.1 Development of multi-dimensional indicator systems for photovoltaic power generation

In photovoltaic (PV) power generation research, an increasing number of scholars have recognized that relying solely on indicators such as “power generation efficiency” or “economic viability” cannot comprehensively assess the sustainability and application potential of PV systems (Liu J. et al., 2025). Early studies incorporated metrics such as energy payback time (Barbosa, 2026), return on investment (Lei et al., 2025), greenhouse gas emission intensity (Xu et al., 2025), and net present value (Ellis et al., 2023) into a unified evaluation framework, thereby enabling the comprehensive ranking of different PV system options. Furthermore, some research has developed a comprehensive evaluation system for the feasibility of PV projects (Djalab et al., 2024), which considers factors such as technology type (Si et al., 2024), cost (Chong et al., 2024), environmental impact (Si et al., 2024), and policy support (Chong et al., 2023). With the advancement of Geographic Information Systems (GIS) and remote sensing technologies, an increasing number of studies have introduced spatial dimensions into site selection and potential assessment (Lambarki et al., 2025), incorporating metrics such as solar radiation (Yao et al., 2024), land use (Sanda et al., 2026), and slope/aspect (Dhivagar et al., 2025) into evaluation systems. This enables the provision of reasonable layout recommendations for both centralized and distributed PV systems. Some studies have attempted to integrate the “lifecycle terminal management” of PV systems into multidimensional evaluation methods (Liu H. et al., 2025). For instance, Hernández-López et al. (2024) assessed the treatment methods of waste PV modules from environmental,

technical, social, regulatory, and economic perspectives, offering a reference for the lifecycle management of the PV industry. Moreover, the analysis of broader energy systems has also provided insights for PV evaluation frameworks: for example, Lia et al. (2025) developed a comprehensive evaluation system covering seven dimensions, including energy efficiency, environmental impact, economic viability, social benefits, and system development potential, for assessing rural integrated energy systems (including PV, wind, and biomass). This method integrates expert weighting and data-driven weighting to balance expertise and objectivity.

Despite the significant progress made in advancing PV evaluation frameworks from simple to complex, notable shortcomings remain. First, many studies still limit their scope to a three-dimensional framework focusing on energy, economy, and environment (Lei et al., 2025), with limited inclusion of social dimensions such as land use conflicts, community acceptance, local economy and employment, environmental carrying capacity, and long-term, external influencing factors like system terminal management (Lambarki et al., 2025; Yao et al., 2024; Sanda et al., 2026). Second, different studies vary greatly in their selection of indicators and weight-setting methods for different systems (e.g., centralized, distributed, agricultural PV systems). For example, some studies rely on expert judgment (Liu H. et al., 2025), while others attempt data-driven methods (Lia et al., 2025), but a lack of standardized approaches makes it difficult to directly compare results across studies. Furthermore, few studies incorporate dynamic factors such as spatial variability, climate, operational stability, and uncertainty into evaluation frameworks, which results in evaluation outcomes better suited to static analysis rather than dynamic realities (Si et al., 2024). Additionally, comprehensive evaluations of the full lifecycle of PV systems remain rare, making it challenging to truly reflect their sustainability (Liu H. et al., 2025).

### 2.2 Impact of photovoltaic power generation on carbon emissions and sustainable development

Over the past decade, numerous studies have demonstrated the significant advantages of photovoltaic (PV) power generation systems over traditional fossil fuel-based power generation in terms of greenhouse gas emissions and long-term sustainability, through methods such as lifecycle assessment and carbon footprint analysis (Aridi et al., 2025). Early research compiled key indicators for different PV technologies, such as monocrystalline silicon (Amrouche et al., 2013), polycrystalline silicon (Amrouche et al., 2013), and thin-film PV (Gu et al., 2025), including energy payback time (Lamnatou et al., 2023) and greenhouse gas emission rates (Lamnatou et al., 2023). The results indicated that the energy payback time for many PV systems typically ranges from 0.75 to 3.5 years, and the greenhouse gas emissions are significantly lower compared to fossil fuel-based power generation (Boujelbene et al., 2026). Subsequently, more studies analyzed the lifecycle assessments of PV systems, reaching a consistent conclusion: whether for crystalline silicon or thin-film PV systems, the greenhouse gas emissions over their lifecycle are significantly lower than those of coal, natural gas, and other traditional energy generation systems

(Hernández-López et al., 2024; Peng et al., 2013; Nugent and Sovacool, 2014).

Recent studies also show that the unit carbon emissions of PV systems in China have significantly decreased from 1.66 kg of CO<sub>2</sub> per watt in 2011 to approximately 0.75 kg by 2018. Furthermore, with every 1% increase in the proportion of PV generation, carbon emissions in China's power sector are expected to decrease by approximately 2.05% by 2035 (Wang et al., 2024). In terms of sustainability assessment, some studies have expanded traditional environmental and economic evaluations by incorporating social impacts, resource utilization, and ecological footprints, as well as introducing uncertainty analysis to address external variables such as climate change, material supply, and policy shifts (Ke et al., 2025a). However, despite the significant carbon reduction benefits of PV systems, most studies have primarily focused on aspects such as greenhouse gas emissions and energy payback time, with less attention given to other potential environmental impacts over the entire lifecycle of PV systems, such as land use changes, ecological effects, and the recycling and disposal of waste PV modules (Dong et al., 2024).

Although the impact of PV power generation on carbon emissions has been widely studied, several gaps remain. First, most studies focus mainly on the direct reduction of carbon emissions, lacking a comprehensive assessment of the environmental impacts at different stages of the PV system lifecycle, particularly during the disposal and recycling phases (Dong et al., 2024). Second, many evaluation methods place too much emphasis on greenhouse gas emissions while overlooking the impacts of PV systems on land use, water resources, and biodiversity (Wang et al., 2024; Ke et al., 2025a).

## 2.3 Methods for photovoltaic power generation forecasting

In recent years, an increasing number of studies have shifted towards using deep learning methods to predict photovoltaic (PV) power generation, addressing the highly fluctuating and uncertain nature of its output due to factors such as weather, meteorological conditions, and geographic variations (Gong et al., 2024). Compared to traditional statistical methods or simple machine learning models, deep neural networks (e.g., Recurrent Neural Networks (RNNs) (Ke et al., 2025b), Convolutional Neural Networks (CNNs) (Raorane et al., 2025), and Transformers (Sun et al., 2025)) have demonstrated significant advantages in capturing complex nonlinear patterns and potential relationships between multiple variables in time series data (Frederiksen and Cai, 2022). For instance, some studies have shown that by training models with historical generation data, meteorological data, and other auxiliary features using CNNs, Long Short-Term Memory Networks (LSTMs), and Discrete Wavelet Transform (DWT), the model can more accurately predict future PV power output (Wei et al., 2025). Other studies have combined CNN, LSTM, and Transformer structures in hybrid models to further enhance prediction accuracy and generalization capability (Dai et al., 2025; Abdel-Basset et al., 2021).

However, these deep learning methods still face several limitations and challenges. First, the performance of these models is highly dependent on high-quality, continuous, and comprehensive historical data. In many real-world PV power

plants, meteorological records and generation data are often incomplete or of inconsistent quality (Sun et al., 2025; Wei et al., 2025). Second, the data preprocessing, feature selection, model architecture, and training/testing partitioning standards vary across different studies, making it difficult to directly compare results between studies (Xia et al., 2025). Currently, there is also a lack of a unified, standardized benchmark dataset and evaluation system (Raorane et al., 2025).

## 2.4 Research innovations

Existing research on photovoltaic power generation has often focused on optimizing individual components, such as improving energy efficiency, reducing costs, or minimizing environmental impact. However, many studies fail to consider the complex interactions between various factors, including environmental conditions, technological advancements, and economic considerations. Additionally, most models rely on simplified assumptions or narrow evaluation criteria, overlooking broader factors like system reliability, regional variations, and long-term sustainability. The innovation in this paper lies in its integrated approach. We combine t-SNE dimensionality reduction with an enhanced particle swarm optimization algorithm to improve the efficiency of solving complex nonlinear problems. By incorporating multiple indicators—such as energy production, environmental impact, and economic feasibility—and introducing a dynamic optimization framework, the proposed model can more accurately predict energy output, optimize system configurations, and adapt to regional and temporal changes. This approach improves the accuracy of photovoltaic system modeling and offers a more comprehensive and sustainable solution to the challenges of solar power generation.

## 3 Construction of power supply forecasting model and selection of indicators

### 3.1 Data source

The dataset used in this study was compiled from official statistical publications and publicly accessible databases for China, with all variables harmonized to an annual frequency before analysis. Specifically, macroeconomic and demographic indicators (e.g., GDP, *per capita* GDP, population density, and urban population share) were collected primarily from the National Bureau of Statistics of China (NBS) and the China Statistical Yearbook. Energy-related indicators (e.g., total energy consumption and the consumption of coal, oil, and natural gas) were obtained from the China Energy Statistical Yearbook. Power-sector and electricity-related indicators (including electricity generation and selected power industry variables) were collected from the China Electric Power Yearbook, relevant publications of the China Electricity Council (CEC), and publicly released statistics from the National Energy Administration (NEA) where applicable. Environmental and emissions-related indicators (including carbon-emission-related variables used in the carbon analysis module) were compiled from official statistical publications and publicly available

emission datasets (e.g., national statistical publications and open-access carbon-emissions databases, depending on variable availability and year coverage). The historical data were used for model estimation and validation, and the projected period (2024–2060) was used only for scenario-based long-term assessment.

Potential data uncertainties arise from missing observations, revisions in statistical yearbooks, and inconsistencies in indicator definitions or reporting standards across different sources and years. Additional uncertainty may also be introduced during preprocessing, especially when outliers are removed and missing values are imputed. To mitigate these effects, we applied a standardized preprocessing workflow including outlier screening, missing-value imputation, and consistency checks prior to model construction. These uncertainties were considered in the interpretation of the results, particularly for long-horizon projections, which should be understood as scenario-based estimates rather than exact point predictions.

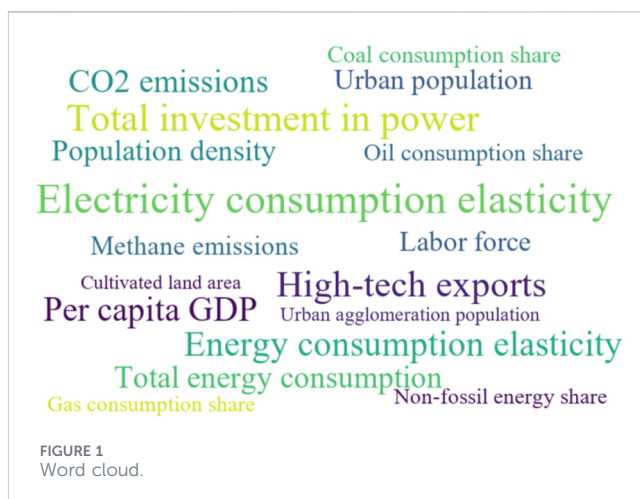
### 3.2 Search for indicators

To construct a multidimensional indicator system for photovoltaic (PV) power generation analysis, we performed a structured literature retrieval using CNKI, PubMed, and Google Scholar. To reduce terminology bias, the search strategy was expanded beyond the single term “power supply” and included semantically related terms such as photovoltaic, PV, solar power generation, and solar cell. The retrieved records were then screened for relevance to PV power generation assessment, forecasting, optimization, and carbon-emission-related analysis. After relevance screening and removal of duplicates, 34 studies were retained for indicator extraction.

In the initial screening stage, indicator-related terms and keywords were extracted from the retained studies and ranked by frequency to identify candidate indicators. However, word-frequency statistics were used only as a preliminary screening tool and not as the sole basis for final indicator selection. To address the limitations of frequency-only classification (e.g., synonymy, polysemy, and context dependence), we introduced a semantic validation procedure.

The semantic validation procedure included three steps. First, synonym normalization and term merging were conducted to unify semantically equivalent expressions (e.g., PV, photovoltaic, and solar power generation) and reduce artificial fragmentation of concept counts. Second, context-based manual screening was performed to verify whether high-frequency terms represented meaningful analytical indicators in the source literature, rather than general descriptors or context-specific expressions unrelated to indicator construction. Third, a thematic grouping (topic-level) cross-check was applied to ensure that the retained indicators reflected the dominant thematic content of the literature corpus rather than isolated term occurrences.

To improve transparency, [Figure 1](#) (word cloud) and [Figure 2](#) (frequency statistics of the top indicators) are retained as descriptive visualizations of the initial extraction stage. These figures illustrate the distribution of candidate terms but do not by themselves determine the final indicator set. The final indicator system was established based on the combined frequency-semantic validation procedure described above.



Ultimately, the evaluation system is constructed based on the following five primary indicators: Economic and Industrial Indicators, Energy Consumption and Supply Indicators, Environmental and Emissions Indicators, Population and Urbanization Indicators, and Sustainability Indicators.

#### 3.2.1 Economic and Industrial Indicators

These indicators are crucial for understanding the economic development and industrial structure of a country or region. Key indicators like investment in the power industry and *per capita* GDP provide insights into the overall development stage and the potential demand for renewable energy, particularly solar power. By analyzing the proportion and importance of sectors such as the power industry and high-tech industries, we can assess the rationality of the industrial structure and its future direction. These indicators help to determine the economic feasibility of scaling up photovoltaic energy production and provide the government with data to formulate effective industrial policies. They also ensure that economic growth aligns with sustainable energy development, fostering a healthy and robust energy transition.

#### 3.2.2 Energy consumption and supply indicators

Energy consumption patterns directly impact the effectiveness and sustainability of photovoltaic power generation. Indicators such as total energy consumption and the energy consumption elasticity coefficient reveal the relationship between energy use and economic growth, indicating how efficiently energy resources, including solar energy, are utilized in driving economic development. Understanding energy supply and demand helps countries and regions strategically plan their energy mix, integrate renewable sources, and make informed decisions about solar energy adoption to meet future demand sustainably.

#### 3.2.3 Environmental and emissions indicators

These are essential for evaluating the environmental impacts of energy consumption and production, especially from traditional fossil fuels. Greenhouse gas emissions, methane emissions, and



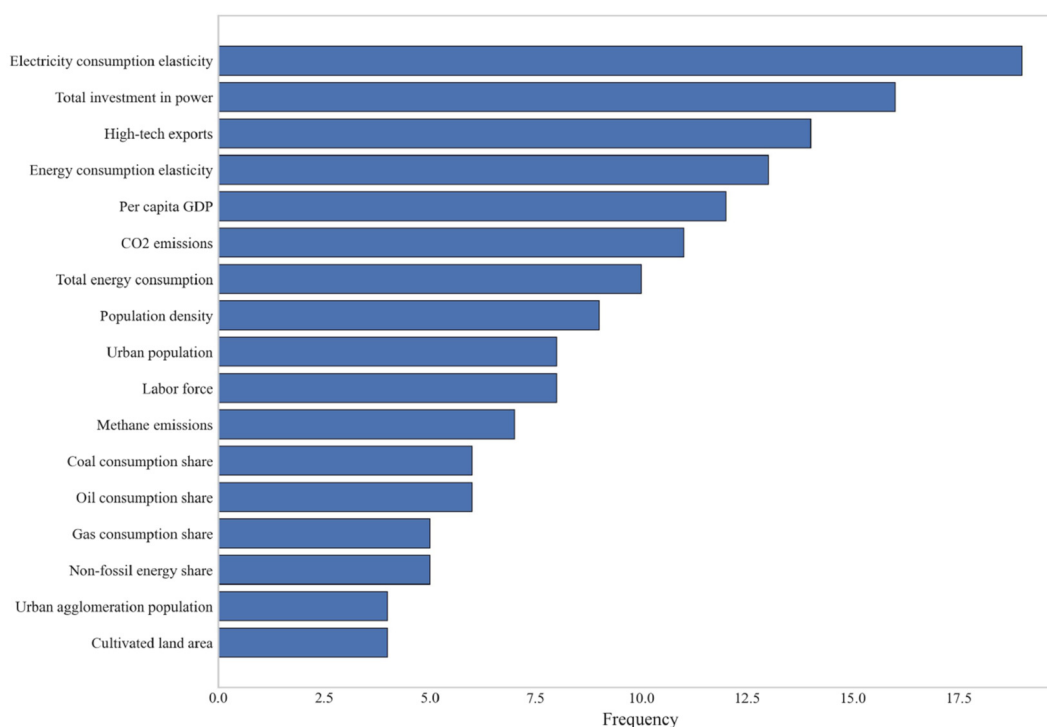


FIGURE 2  
Frequency statistics bar chart.

carbon dioxide emissions are directly related to the environmental impact of power generation systems. These indicators help assess the effectiveness of policies aimed at reducing carbon emissions and mitigating climate change. They also guide the formulation of environmental protection strategies, ensuring that photovoltaic power generation contributes positively to environmental sustainability by reducing carbon footprints compared to fossil-fuel-based energy sources.

### 3.2.4 Population and urbanization indicators

These indicators, such as population density and urban population share, are linked to the demand for energy and social services. Urbanization often leads to higher energy demand, making it crucial to include these indicators in the study. Understanding population growth and distribution helps in planning for energy demand, ensuring that solar power systems are deployed efficiently in urban and rural areas alike. This also aids in optimizing energy distribution and enhancing social service delivery.

### 3.2.5 Sustainability indicators

Finally, sustainability indicators such as cultivated land area and the efficiency of energy use provide important insights into the long-term viability of energy production systems. These indicators are particularly useful for assessing the potential for large-scale photovoltaic deployment without negatively impacting land use or natural resources. By measuring the sustainability of photovoltaic energy alongside economic, social, and environmental factors, we ensure that the development of solar

power aligns with sustainable development goals and contributes to a greener, more sustainable energy future.

Together, these indicators provide a comprehensive framework for evaluating photovoltaic power generation's potential and its role in achieving sustainable energy transitions, minimizing environmental impacts, and fostering economic development.

## 3.3 Data preprocessing - outliers

The Kolmogorov-Smirnov test is a non-parametric statistical test, a statistical method used to test whether a dataset follows a certain distribution, most commonly to test whether a dataset follows a normal distribution. The basic principle is to compare the cumulative distribution function of the dataset with the theoretical distribution function and determine whether the dataset conforms to the theoretical distribution by calculating the maximum gap between the two. If the maximum gap is less than a critical value, the dataset is considered to follow the theoretical distribution, and the single-sample K-S test is used to examine whether the observed empirical distribution of a data is the known theoretical distribution. When the gap between the two is very small, it is inferred that the sample is taken from a known theoretical distribution. The theoretical distribution that serves as the null hypothesis is typically a one-dimensional continuous distribution  $F$  (such as the normal distribution, uniform distribution, exponential distribution, etc.) and is sometimes used for discrete distributions (such as the Poisson distribution). That is  $H$ : Population  $X$  follows some kind of one-dimensional continuous distribution  $F$  (Nie et al., 2024). The test statistic is (Equation 1):

$$Z = \sqrt{n} \max_i (|F_n(x_{i-1}) - F(x_i)|, |F_n(x_i) - F(x_i)|) \quad (1)$$

here,  $Z$  converges to the Kolmogorov-Smirnov distribution. That is, when the sample is taken from a one-dimensional continuous distribution  $F$  (Equation 2):

$$Z \rightarrow K = \sup |B(F(x))| \quad (2)$$

where,  $F$  is a continuous distribution, the distribution of the random variable  $K$  does not depend on  $F$ . The result of the KS test is usually a p-value, and if the p-value is less than the significance level (generally 0.05), the null hypothesis is rejected, that is, the two samples are considered to come from different distributions (Zheng et al., 2025). This paper requires the use of SPSS to plot Q-Q plots for some of the data and to conduct the Kolmogorov-Smirnov test to determine the distribution pattern. The Kolmogorov-Smirnov test analysis results are shown in the following Figure 3:

From the Q-Q plot, most points generally follow the reference line, especially around the median quantiles, suggesting that the distribution of *per capita* GDP is approximately normal. Deviations are observed at both tails, which is common in real-world data. To provide a more objective assessment, a Kolmogorov-Smirnov (K-S) test was further conducted, and the results are presented in Table 1 (Fasano and Franceschini, 1987).

The p-value of the Shapiro-Wilk test is very close to 0 for all 18 columns of data, meaning that the null hypothesis of a normal distribution can be statistically rejected. In other words, none of the 18 columns of data are likely to come from a normal distribution.

To avoid situations where the actual data do not conform to the normal distribution model, a box plot is introduced in this paper. The box plot theory does not impose restrictions on the data itself, is not affected by outliers, can directly and intuitively describe the discrete distribution of the data, and provides a criterion for identifying outliers. The box plot is a method of describing data using the minimum, first quartile, median, third quartile, and maximum values of the data, which does not require the data to follow a normal distribution (Hussain et al., 2021).

Sort the test data in ascending order to get an ordered sequence, and the median is denoted as:

For odd  $n$ , the expression is:

$$M = X_{n+1/2}$$

For even  $n$ , the expression is:

$$M = \frac{1}{2} \left( \frac{X_{n/2}}{2} + X_{(1+n/2)} \right)$$

Outliers are judged by the criteria

$$X_i > U + K \cdot IQR \mid X_i < L - K \cdot IQR$$

In the formula:  $U$ : The upper quartile point ( $Q_3$ ), which is the median of the upper half of the sample. It indicates that only 14 values in the sample are greater than  $U$ .

$L$ : The lower quartile point ( $Q_1$ ), which is the median of the lower half of the sample. It indicates that only 14 values in the sample are smaller than  $L$ .

$IQR$ : The interquartile range, calculated as  $IQR = U - L$ .

$K$ : The step coefficient, typically set to  $K = 1.5$ .

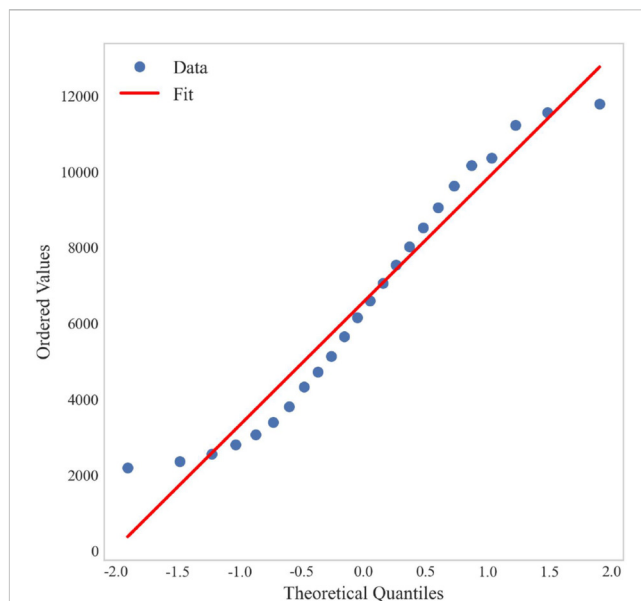


FIGURE 3  
q-q plot of GDP *per capita*.

During data preprocessing, we first used a box plot to screen for outliers and removed the identified abnormal values to improve data quality (Figure 4). Missing values caused by outlier removal or incomplete records were subsequently imputed using interpolation methods.

### 3.4 Data preprocessing - missing values

Missing values were identified after data import. The missing data mainly originated from two sources: 1. values removed during outlier processing and 2. incomplete observations in the original dataset. These missing values were imputed using interpolation methods to avoid potential bias in subsequent analyses. For variables with consecutive missing values (including those associated with outlier-affected segments) that were important for the subsequent analysis, Newton interpolation was applied to reconstruct the missing observations (Al Din, 2020).

The calculation formula is:

$$\begin{aligned} \varepsilon_n(x) = & f(x_0) + (x - x_0)f[x_0, x_1] + (x - x_0)(x - x_1)f[x_0, x_1, x_2] \\ & + \cdots + (x - x_0)(x - x_1) \cdots (x - x_n)f[x_0, x_1, \dots, x_n] \end{aligned}$$

where:  $f[x_0, x_1, \dots, x_n]$  represents the divided difference (the difference quotient), defined as:

$$f[x_0, x_1, \dots, x_n] = \frac{f[x_0, x_1, \dots, x_{n-1}] - f[x_0, x_1, \dots, x_n]}{x - x_n}$$

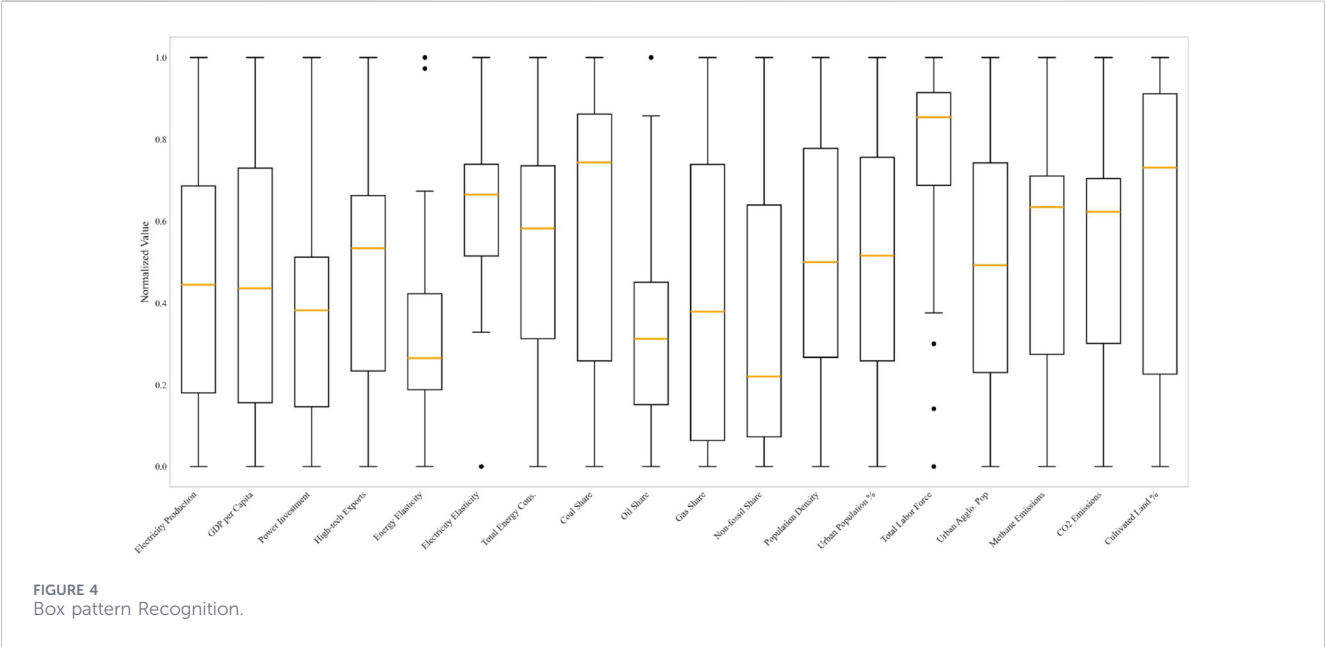
In the formula,  $n$  is the number of missing items, representing the  $i$ th missing item.  $x_i$  Using this formula, some of the results were obtained as follows Figure 5.

### 3.5 Data preprocessing - dimensionality reduction

First, perform the KMO test and the sphericity test to determine the correlation of these secondary indicators under

TABLE 1 Partial results of the K-S test.

| Indicators                                                                          | P value | Metrics                                                                  | P value |
|-------------------------------------------------------------------------------------|---------|--------------------------------------------------------------------------|---------|
| Power production (billion kilowatt-hours)                                           | 0       | Natural gas as a share of total energy consumption (%)                   | 0       |
| Per capita GDP (2015 constant price US dollars)                                     | 0       | Share of total energy from primary electricity and other sources (%)     | 0       |
| Total investment in the production and supply of electricity (in ten thousand yuan) | 0       | Population density (number of people per kilometer of land area)         | 0       |
| High-tech exports (current price US dollars)                                        | 0       | Urban population (proportion of the total population)                    | 0       |
| Energy consumption elasticity coefficient (–)                                       | 0       | Labor force, total                                                       | 0       |
| Electricity consumption elasticity coefficient                                      | 0       | Population in urban agglomerations with a population of over one million | 0       |
| Total energy consumption (tons of standard coal)                                    | 0       | Methane emissions (thousand tons of carbon dioxide equivalent)           | 0       |
| Coal as a share of total energy consumption (%)                                     | 0       | Carbon dioxide emissions (thousand tons)                                 | 0       |
| Proportion of oil in total energy consumption (%)                                   | 0       | Cultivated land (percentage of land area)                                | 0       |



the primary indicators, and then select the dimensionality reduction method. Here, principal component analysis is mainly used for dimensionality reduction of multi-dimensional indicators. Generally speaking, for principal component analysis, the greater the connection between indicators, the better the dimensionality reduction effect. Therefore, this paper uses the KMO test and the Bartlett sphericity test to validate the indicators in advance and determine the relationship between the indicators. For metrics that have not undergone the KMO test and Bartlett’s sphericity test, this paper uses the t-SNE method to reduce the dimensionality of multi-dimensional nonlinear metrics to a two-dimensional sequence, thereby achieving the purpose of dimensionality reduction. Among them, the KMO test and the Bartlett’s sphericity test are used to determine the collinearity or correlation between the collected indicators (Abdi et al., 2010).

$$KMO = \frac{\sum \sum_{i \neq j} r_{ij}}{\sum \sum_{i \neq j} r_{ij}^2 + \sum \sum_{i \neq j} a_{ij}^2}$$

where,  $r_{ij}$  represents the simple correlation coefficient and  $a_{ij}^2$  represents the partial correlation coefficient.

The results of the numerical processing were imported into SPSS for the multi-dimensional index KMO test and Bartlett’s sphericity test as shown below Table 2.

From the test results, it can be seen that the KMO value for the economic and industrial category is 0.659, indicating that factor analysis is generally not very suitable. The KMO value for the energy consumption and structure class is 0.5, which is on the boundary between less suitable and suitable, and may require careful consideration of whether to conduct factor analysis. The Population and Society class has a KMO value of 0.784, indicating a better fit for factor analysis. The KMO value for the

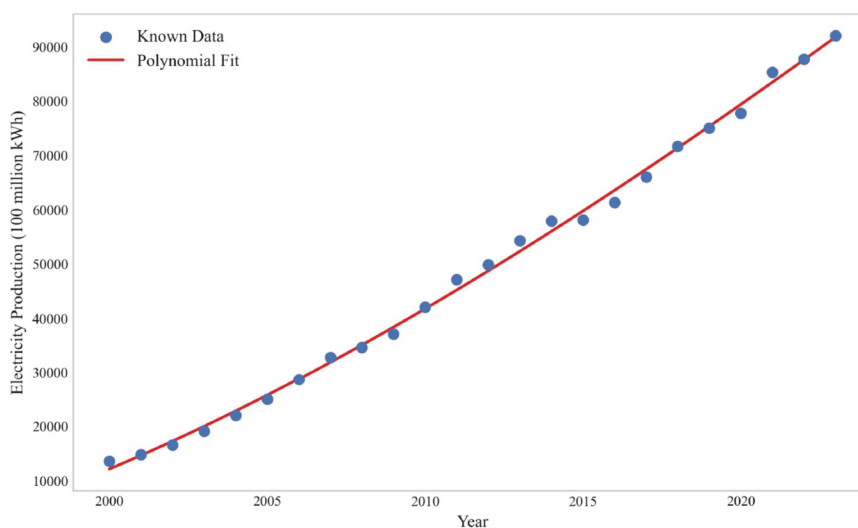


FIGURE 5  
Results after processing.

TABLE 2 Results of KMO test and Bartlett's sphericity test.

| Index names              |                        | Economics and industry | Energy consumption and structure | Population and society | Environment and emissions |
|--------------------------|------------------------|------------------------|----------------------------------|------------------------|---------------------------|
| KMO values               |                        | 0.659                  | 0.5                              | 0.784                  | 0.656                     |
| Bartlett sphericity test | Approximate chi-square | 127.077                | 389.177                          | 278.761                | 130.094                   |
|                          | df                     | 3                      | 21                               | 6                      | 3                         |
|                          | p                      | 0.000*                 | 0.000*                           | 0.000*                 | 0.000*                    |

\* Represent significance levels of 1%, 5%, and 10%, respectively.

environment and emissions category is 0.656, which is also not very suitable for factor analysis.

For population and society, this paper uses principal component analysis for dimensionality reduction, and for energy consumption and structure, t-SNE for dimensionality reduction. For the population and society category and the environment and emissions category, since the number of indicators is limited, this paper directly uses the raw data for analysis.

The main steps for dimensionality reduction through principal component analysis:

1. Standardization: Since the data indicators have different units (e.g., weight in kilograms and rice intake in units), it's important to standardize the data to eliminate unit discrepancies. The formula for standardizing each collected value  $a_{ij}$  is:

$$\tilde{a}_{ij} = \frac{a_{ij} - \mu_j}{s_j}$$

where:  $a_{ij}$  is the value of the  $i$ -th sample for the  $j$ -th indicator.  $\mu_j$  is the mean of the  $j$ -th indicator.  $s_j$  is the standard deviation of the  $j$ -th indicator.

2. Correlation Coefficient Matrix: To capture the relationships between different indicators, calculate the correlation coefficients between each pair of indicators and construct the correlation coefficient matrix  $R = (r_{ij})$ . The formula for calculating the correlation coefficient  $r_{ij}$  between the  $i$ -th and  $j$ -th indicators is:

$$r_{ij} = \frac{\sum_{k=1}^n \tilde{a}_{ki} \cdot \tilde{a}_{kj}}{n-1}$$

where:  $n$  is the total number of samples.  $\tilde{a}_{ki}$  and  $\tilde{a}_{kj}$  are the standardized values for the  $i$ -th and  $j$ -th indicators, respectively.

3. Calculate Eigenvalues and Eigenvectors: Using the correlation matrix  $R$ , calculate the eigenvalues  $\lambda_1, \lambda_2, \dots, \lambda_m$  and the corresponding eigenvectors. The eigenvalues represent the variance captured by each component. The eigenvalues are ordered as follows:

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_m \geq 0$$

where  $m$  is the total number of indicators.

4. Select Principal Components: The contribution rate and cumulative contribution rate are calculated based on the eigenvalues from Step 3. You can then choose the most



TABLE 3 Explanation of total variance.

| Components | Feature roots |                                  |                                          |
|------------|---------------|----------------------------------|------------------------------------------|
|            | Feature roots | Variance interpretation rate (%) | Cumulative variance explanation rate (%) |
| 1          | 3.518         | 87.938                           | 87.938                                   |
| 2          | 0.48          | 12.002                           | 99.94                                    |
| 3          | 0.002         | 0.048                            | 99.988                                   |
| 4          |               | 0.012                            | 100                                      |

important principal components (those with higher contribution rates) and discard those with lower contribution rates. The formulas for the contribution rate  $g_j$  and cumulative contribution rate  $l_j$  are:

Contribution Rate:

$$g_j = \frac{\lambda_j}{\sum_{k=1}^m \lambda_k}$$

Cumulative Contribution Rate:

$$l_j = \frac{\sum_{k=1}^j \lambda_k}{\sum_{k=1}^m \lambda_k}$$

where:  $\lambda_j$  is the eigenvalue of the  $j$ -th principal component.  $g_j$  is the contribution rate of the  $j$ -th principal component.  $l_j$  is the cumulative contribution rate.

5. Calculate the Overall Score: Finally, project each sample onto the new coordinate system formed by the selected principal components. The principal component score  $Z$  for each sample is calculated using the contribution rates of the principal components as weights:

$$Z = \sum_{j=1}^p g_j y_j$$

where:  $g_j$  is the contribution rate of the  $j$ -th principal component.  $y_j$  is the value of the  $j$ -th principal component score.

The specific results of calculating the eigenvalues based on Step 3 are as follows Table 3:

For a more intuitive presentation of the results, a lithotripsy plot was drawn for visual representation Figure 6.

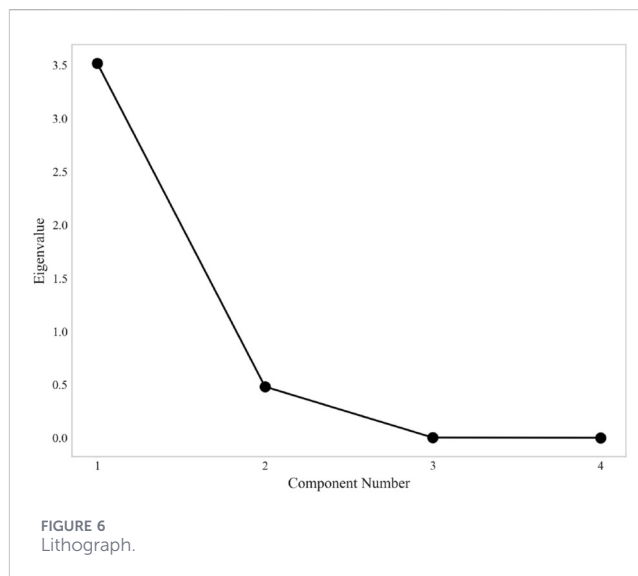
Based on the results of the lithotripsy plot, this paper selects two principal components and obtains the factor loading coefficient table as shown below Table 4.

Based on the combined results of the relevant plots, the first two principal components were selected as the new indicators. To improve the interpretability of these components, a cumulative explained variance plot is provided in Figure 7.

The final dimensionality reduction results are as follows Table 5:

For t-SNE dimensionality reduction, the original data were reduced to two dimensions, and the results are visualized in Figure 8.

Finally, the dimensionality reduction results for the problem are obtained Table 6:



### 3.6 Relational models

Covariance measures the joint variability of two random variables. A positive covariance means that as one variable increases, the other tends to increase as well (and vice versa for a negative covariance). The formula for sample covariance between two variables  $X$  and  $Y$  is:

$$\text{Cov}(X, Y) = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$

where:  $X_i$  and  $Y_i$  are individual data points of  $X$  and  $Y$ ,  $\bar{X}$  and  $\bar{Y}$  are the sample means of  $X$  and  $Y$ ,  $n$  is the number of samples.

The sample standard deviation measures the dispersion of data points around the mean. The formula for the sample standard deviation of a variable  $X$  is:

$$S_X = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1}}$$

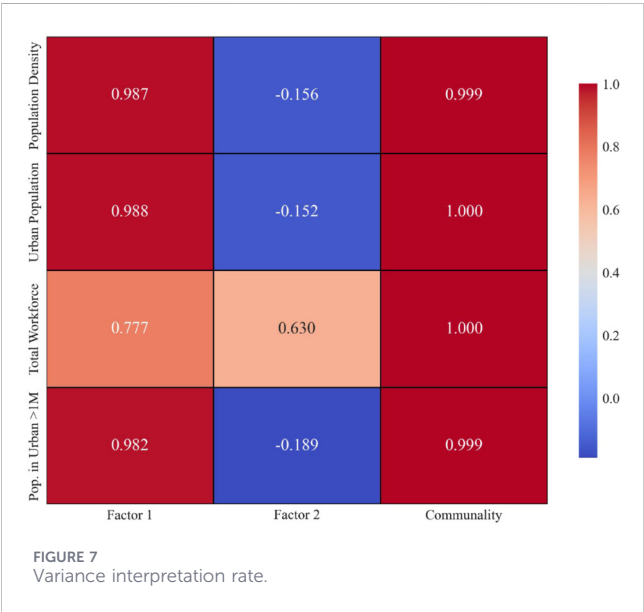
where:  $X_i$  represents each data point,  $\bar{X}$  is the mean of the variable,  $n$  is the number of data points.

The Pearson correlation coefficient  $r_{xy}$  measures the strength and direction of the linear relationship between two variables. It is calculated as:

$$r_{xy} = \frac{\text{Cov}(X, Y)}{S_X S_Y}$$

TABLE 4 Factor loading coefficient table.

| Variables                                                                     | Factor loading coefficients |                       | Commonality (common factor variance) |
|-------------------------------------------------------------------------------|-----------------------------|-----------------------|--------------------------------------|
|                                                                               | Principal component 1       | Principal component 2 |                                      |
| Population density (number of people per kilometer of land area)              | 0.987                       | −0.156                | 0.999                                |
| Urban population (proportion of total population)                             | 0.988                       | −0.152                | 1                                    |
| Labor force, total number                                                     | 0.777                       | 0.63                  | 1                                    |
| Population in urban agglomerations with a population of more than one million | 0.982                       | −0.189                | 0.999                                |



where:  $Cov(X, Y)$  is the sample covariance between  $X$  and  $Y$ ,  $S_X$  and  $S_Y$  are the sample standard deviations of  $X$  and  $Y$ , respectively.

The calculation results are shown in Figure 9.

Figure 9 (Pearson correlation heatmap) shows a clear and highly structured correlation pattern among the variables. Electricity production, population and society, GDP *per capita*, power investment, high-tech exports, and methane/CO2 emissions are all strongly and positively correlated with one another (most coefficients are around 0.90 or higher), indicating strong co-movement among economic development, energy activity, and emission-related indicators over the sample period. In particular, methane emissions and CO2 emissions exhibit an almost perfect positive correlation, suggesting that the two emission indicators follow very similar trends. In contrast, cultivated land and Energy Cons.1 are strongly negatively correlated with most of the above variables, while cultivated land is positively correlated with Energy Cons.1 (approximately 0.89), implying a different variation pattern associated with land use and part of the energy-consumption structure. Energy Cons.2 shows moderate-to-strong positive correlations with economic and emission-related variables (about 0.77–0.81), but negative correlations with cultivated land and Energy Cons.1. Overall, the

results indicate substantial linear dependence and potential multicollinearity among the variables, which provides statistical support for the subsequent use of dimensionality reduction methods (e.g., PCA) to extract representative features.

### 3.7 The establishment and solution of the predictive model

Cubic polynomial regression model is a statistical technique used to describe the relationship between variables, especially when the data points show a nonlinear trend (Ellis et al., 2023). In this case, this paper uses cubic polynomials, which means the highest power of the independent variable (such as the year) is 3. The basic form of a cubic polynomial regression model is as follows:

$$y = a + bx + cx^2 + dx^3 + \epsilon$$

Among them:  $y$  is the dependent variable (e.g., electricity production).  $x$  is the independent variable (for example, the year).  $a, b, c, d$  are model parameters that represent the intercept and slope of the regression equation.  $\epsilon$  is the error term, representing the part of the variable that the model cannot explain.

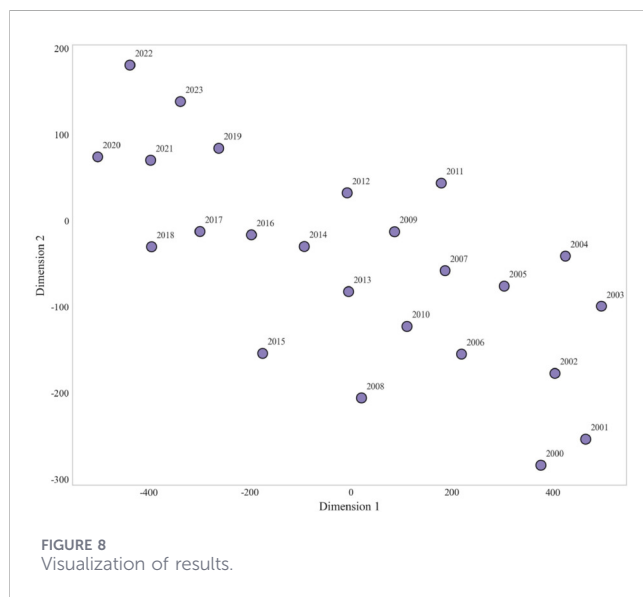
In cubic polynomial regression, by adjusting parameters  $a, b, c, d$ , the model attempts to minimize the difference between the actual value  $y$  and the predicted value, typically by minimizing the sum of squared errors.

The main difference between cubic polynomial regression and linear regression is that it allows the model to capture more complex relationships. For example, in the case of electricity production, if the data shows a curve rather than a linear trend, then using polynomial regression can fit and predict the data more accurately Figure 10.

Figure 10 shows the fitted results of the cubic polynomial regression model for historical electricity production and the long-term projection to 2060, together with the 95% prediction interval. The model provides a close fit to the observed data during the historical period, indicating that the cubic form can capture the nonlinear growth trend in electricity production. However, as the forecast horizon extends beyond the observed period, the prediction interval widens progressively, forming a clear “cone of uncertainty.” This pattern indicates increasing uncertainty in long-term extrapolation and suggests that prediction reliability decreases with time.

TABLE 5 Dimensionality reduction results table.

| Year | Population and society category | Year | Population and society category |
|------|---------------------------------|------|---------------------------------|
| 2000 | -1.94381776                     | 2012 | 0.25775141                      |
| 2001 | -1.670738222                    | 2013 | 0.351489115                     |
| 2002 | -1.382909057                    | 2014 | 0.466545981                     |
| 2003 | -1.108143532                    | 2015 | 0.570652252                     |
| 2004 | -0.850977287                    | 2016 | 0.654484495                     |
| 2005 | -0.589247538                    | 2017 | 0.699689608                     |
| 2006 | -0.382301203                    | 2018 | 0.724356656                     |
| 2007 | -0.253358762                    | 2019 | 0.777889098                     |
| 2008 | -0.158115185                    | 2020 | 0.265584376                     |
| 2009 | -0.089471442                    | 2021 | 1.032102083                     |
| 2010 | -0.038582232                    | 2022 | 1.188232345                     |
| 2011 | 0.151803056                     | 2023 | 1.327081744                     |

FIGURE 8  
Visualization of results.

### 3.8 Model evaluation metrics

In order to test the prediction results of the prediction model, this paper uses root mean square error (RMSE), mean absolute percentage error (MAPE), and determination coefficient ( $R^2$ ) as evaluation indicators for establishing the time-based economic prediction model in this paper.

The RMSE measures the average magnitude of the error between predicted values  $\hat{y}_i$  and actual values  $y_i$ . It's calculated as:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2}$$

where:  $\hat{y}_i$  is the predicted value for the  $i$ -th observation,  $y_i$  is the actual value for the  $i$ -th observation,  $n$  is the total number of predictions.

The MAPE expresses the average absolute percentage difference between predicted values and actual values:

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{\hat{y}_i - y_i}{y_i} \right| \times 100\%$$

where:  $\hat{y}_i$  is the predicted value for the  $i$ -th observation,  $y_i$  is the actual value for the  $i$ -th observation,  $n$  is the total number of observations.

The  $R^2$  value measures how well the model explains the variability of the data. It is calculated as:

$$R^2 = 1 - \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

where:  $\hat{y}_i$  is the predicted value for the  $i$ -th observation,  $y_i$  is the actual value for the  $i$ -th observation,  $\bar{y}$  is the mean of the actual values,  $n$  is the total number of observations.

To reduce the risk of overfitting and evaluate model generalizability, this study further introduces an out-of-sample validation procedure using five-fold blocked cross-validation (Blocked K-Fold). Unlike standard K-fold cross-validation, the temporal order of the data is preserved (i.e., no random shuffling is applied). The time-ordered dataset is divided into five consecutive folds, and in each iteration one fold is used as the validation set while the remaining four folds are used as the training set. This process is repeated until each fold has been used once as the validation set, and the final validation performance is summarized using the average RMSE, MAPE, and  $R^2$  across the five folds.

Compared with a single holdout split, blocked K-fold validation makes more efficient use of the available data and provides a more stable performance estimate. However, because some folds may use observations both earlier and later than the validation block for training, this approach does not represent a strictly forward-looking forecasting protocol. Therefore, in this study, blocked K-fold cross-validation is used as a robustness check of model generalization to the

overall data structure rather than a strict real-time forecasting test.

In this study, the in-sample evaluation metrics are RMSE = 122.008, MAPE = 2.64%, and  $R^2 = 0.9975$ . To assess out-of-sample performance, the five-fold blocked cross-validation results are also reported, yielding RMSE = 168.35, MAPE = 5.56%, and  $R^2 = 0.9236$ . The lower performance in cross-validation relative to the in-sample fit is expected and indicates that the model's predictive capability is weaker under out-of-sample evaluation. Nevertheless, the cross-validation results suggest that the model still captures the main trend of electricity production reasonably well.

# 4 Construction of photovoltaic power generation optimization model and solution using particle swarm optimization algorithm

## 4.1 Model construction

To support long-term PV planning under uncertainty, this study develops a scenario-based optimization model for photovoltaic (PV) power generation. The model is designed to maximize total power generation over the planning period subject to economic and technical constraints, including investment and operation costs, grid-capacity limits, budget restrictions, and feasible parameter ranges. Given the uncertainty associated with long-term projections, several necessary assumptions are specified to define the planning horizon and parameter intervals for model implementation. The objective function and constraints are presented below, followed by the assumptions adopted in the optimization process.

$$\begin{aligned} \text{Max } E_{\text{total}} &= \sum_{i=\text{strat}}^{\text{end}} C_i \times E_i \times R_i \times H \\ \text{s.t. } \left\{ \begin{array}{l} I_i = C_i \times (\cos t_{\text{panel}} + \cos t_{\text{install}} + \cos t_{\text{grid}}) \\ O_i = C_i \times \cos t_{\text{O\&M}} \\ \text{TotalCost}_i = I_i + O_i \\ G_i \leq \text{GridCapacity}_i \\ I_i \leq \text{Budget}_i \\ E_i \in R_E \\ R_i \in K_E \\ C_i \in C_R \end{array} \right. \end{aligned}$$

Based on the solution of this model, the following necessary assumptions are made in this paper: the starting year is 2024 and the ending year is 2060; The initial installed capacity is 100 megawatts (MW), and the annual growth rate of installed capacity is randomly between 1% and 5%; Generation efficiency varies between 15% and 25% per year; Solar radiation varies between 1,500 and 2,000 per year; The annual effective power generation hours are assumed to be 1,825 h (that is, an average of 5 h per day). When optimizing for multiple regions, each region has different solar radiation and installed capacity growth rates, taking into account factors such as installation costs, maintenance costs, and government subsidies. Generation efficiency increases over time to reflect technological progress, and grid capacity constraints are incorporated. Through optimization, the results show the optimal installed capacity for each region each year, and

TABLE 6 Dimensionality reduction results table.

| Energy consumption and structure class 1 | Energy consumption and structure 2 |
|------------------------------------------|------------------------------------|
| 376.1610225                              | −283.5553864                       |
| 465.0131749                              | −253.3858928                       |
| 404.0502685                              | −177.1177761                       |
| 496.0055254                              | −99.42840915                       |
| 425.0238944                              | −40.66218434                       |
| 303.3486727                              | −75.40841747                       |
| 218.9275305                              | −154.9208819                       |
| 186.7202874                              | −57.60970559                       |
| 20.92087749                              | −205.6559987                       |
| 86.68930607                              | −12.39113821                       |
| 110.8208233                              | −122.6809017                       |
| 179.0133014                              | 43.73163528                        |
| −7.360209356                             | 32.52049868                        |
| −4.461561869                             | −82.01600262                       |
| −92.42014416                             | −29.51906145                       |
| −174.7913649                             | −153.8899587                       |
| −196.9360852                             | −16.24932497                       |
| −299.1129216                             | −12.36372099                       |
| −394.8138124                             | −29.95788886                       |
| −261.7572259                             | 84.31428551                        |
| −501.3047177                             | 74.31436159                        |
| −396.8407021                             | 70.43538663                        |
| −437.8071451                             | 180.7220897                        |
| −337.907139                              | 138.3694036                        |

the total revenue is the sum of all regions and years. To improve the accuracy of the algorithm, this paper introduces the particle swarm optimization algorithm, which was first proposed by Eberhart and Kennedy in 1995 and is characterized by simplicity, fast convergence, and few set parameters, which precisely make up for the shortcomings of the model established in this paper. Based on the traditional particle swarm algorithm, this paper improves it in combination with this problem. Through the planning model, each particle determines the individual optimal solution, and from these individual optimal solutions, a global optimal value is found. This process involves the position and distance formulas of the particles (Li et al., 2025).

$$\begin{aligned} z_{ij} &= z_{ij} + d_{1ij} \times \text{rand} \times (pbest_{ij} - x_{ij}) \\ &\quad + d_{2ij} \times \text{rand} \times (pbest_{ij} - x_{ij}) \\ x_{ij} &= x_{ij} + z_{ij} \end{aligned}$$

Here, *rand* is a random number between 0 and 1.

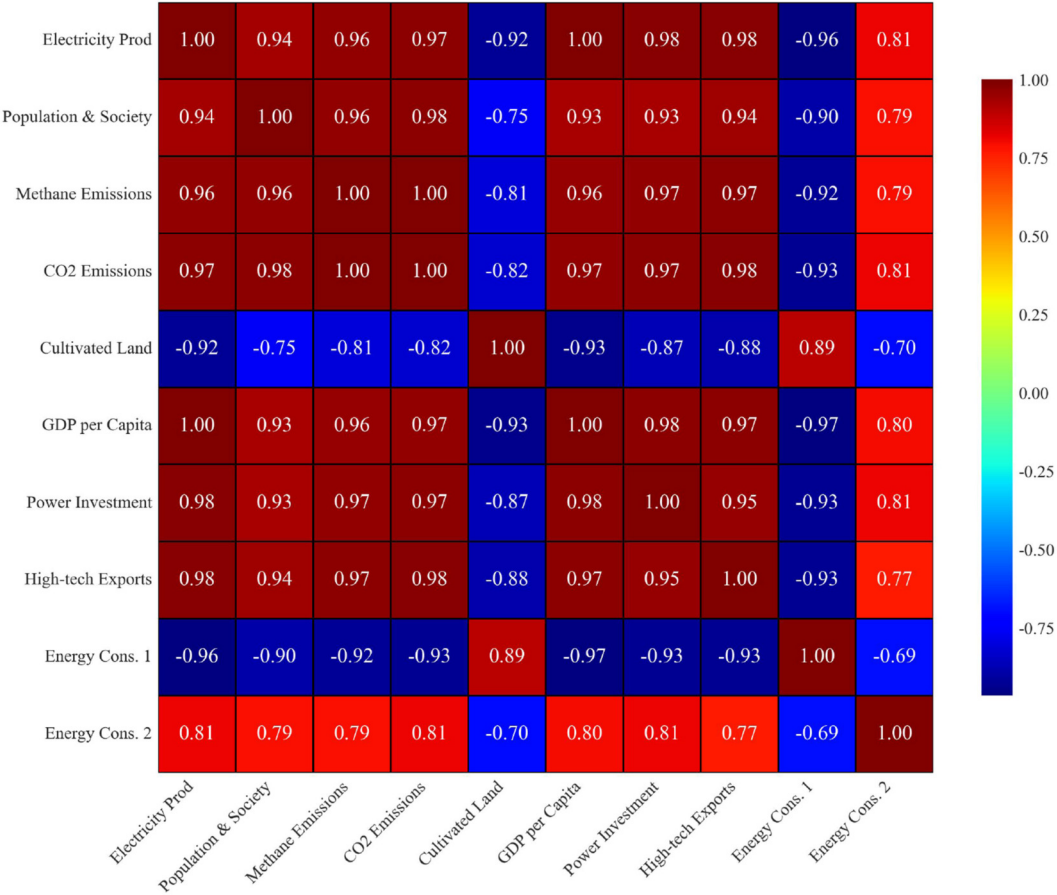


FIGURE 9  
Results of the correlation analysis.

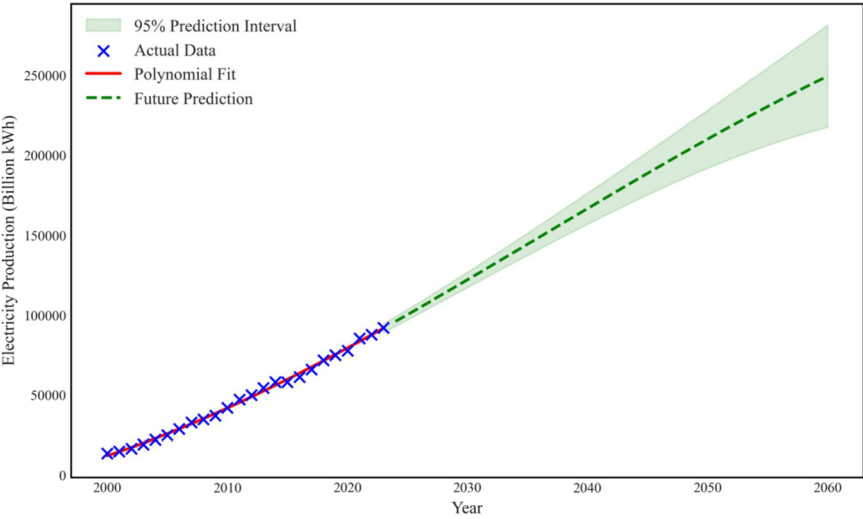


FIGURE 10  
Graph of the results of cubic linear regression predictions.



$$z_{ij} = \omega \times z_{ij} + d_{1ij} \times \text{rand} \times (pbest_{ij} - x_{ij}) + d_{2ij} \times \text{rand} \times (pbest_{ij} - x_{ij})$$

$\omega^{(t)}$  is the inertia factor whose value is non-negative. The larger the value, the stronger the ability to find the optimal solution, and the weaker the ability to find the local optimal solution; conversely, the conclusion is the opposite. In this paper, the strategy of linearly decreasing weights is employed.

$$\omega^{(t)} = (\omega_{ini} - \omega_{end})(G_k - g) / G_k + \omega_{end}$$

$\omega_{ini}$  is the initial inertia factor,  $\omega_{end}$  is the final inertia factor,  $G_k$  is the maximum value (possibly the maximum number of iterations or generations),  $g$  is the current generation or iteration number.

## 4.2 Optimization results for photovoltaic power generation based on particle swarm optimization

“Total power generation: 452,8380,232.71 kWh” is the result calculated by the optimized model under specific constraints and assumptions, representing the total projected cumulative power generation over all considered years and regions. This figure reflects the scale and efficiency of the photovoltaic installations. A higher total power generation during the simulation period from 2024 to 2060 indicates that the installations have a larger installed capacity, higher power generation efficiency, or sufficient solar radiation. The result is influenced by a number of factors, including the amount of solar radiation in each region, the efficiency of photovoltaic panels, the annual growth rate of installed capacity, and the number of effective hours of power generation each year, which are set to directly affect the level of total power generation.

In practical applications, the influences of multiple aspects such as technological progress, economic feasibility and environmental and social factors need to be taken into consideration. Technological advancements are expected to enhance the efficiency of future photovoltaic installations, thereby increasing total power generation; Economic feasibility, however, depends on factors such as cost, market price and policy support, and economic infeasibility may limit its development even if it is technically capable of generating large amounts of electricity. In addition, factors such as environmental impact, land use and social acceptance need to be taken into account when building and operating photovoltaic power stations.

The results suggest that the adopted optimization approach is effective in finding strategies to increase total power generation, but further analysis and validation of other aspects of the model, such as economic benefits and environmental impact, are needed to draw more practical and comprehensive conclusions. In conclusion, this result provides an estimate of the possible total power generation, but in the actual energy planning and decision-making process, more actual circumstances and external factors need to be combined for comprehensive consideration.

## 5 Carbon emission prediction and optimization model construction and sustainable development strategy analysis

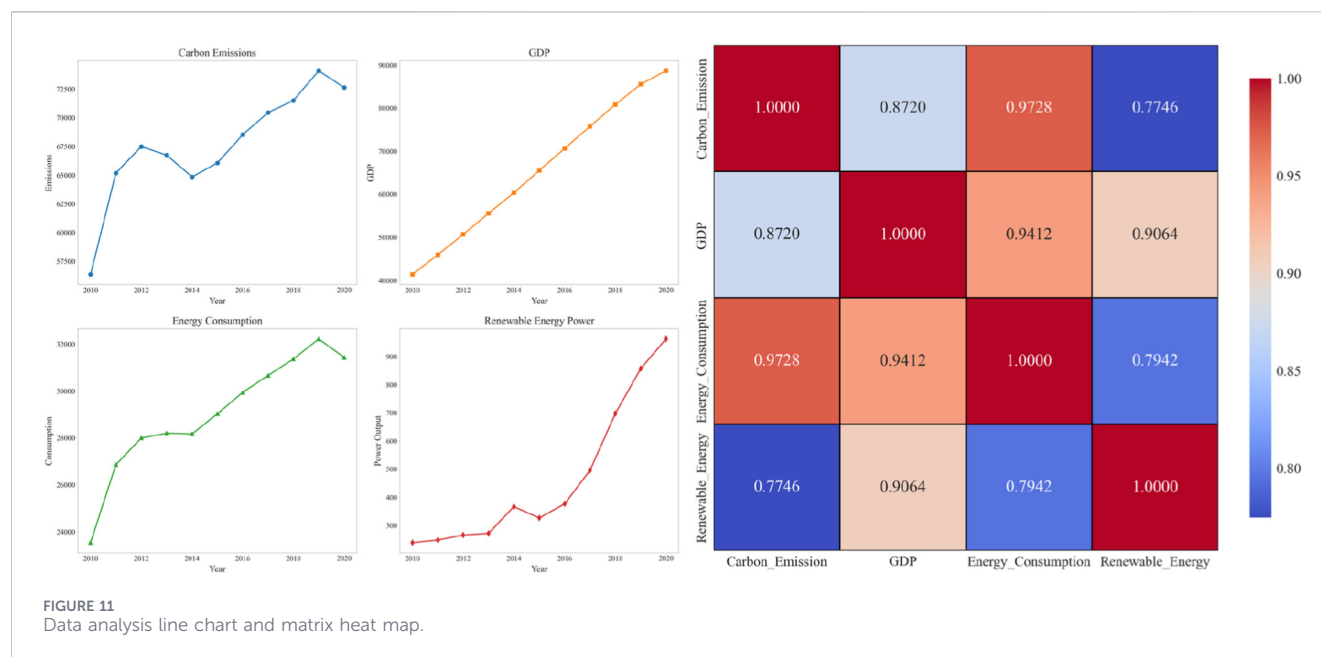
### 5.1 Data organization

For a more systematic analysis of the impact of photovoltaic power generation on carbon emissions, this paper collected data on carbon emissions and related data in a certain area, including GDP, energy consumption, household consumption, coal consumption, oil consumption, natural gas consumption, and data related to new energy power. These indicators are important for assessing regional carbon emissions and formulating emission reduction strategies.

GDP, as a measure of the size of the economy and the total volume of economic activity, is often closely associated with an increase in energy consumption and carbon emissions and is therefore a key factor in analyzing the relationship between carbon emission trends and economic development. Energy consumption directly affects carbon emissions, and by analyzing energy consumption, one can understand the energy demand and structure of a region, thereby identifying potential for emission reduction. Residents' living consumption covers energy consumption such as electricity, gas, transportation, etc. As living standards improve, its contribution to carbon emissions is gradually increasing. Analyzing its trends and composition is crucial for formulating emission reduction measures. The size of coal consumption directly affects carbon emissions, and reducing coal consumption is an important way to lower carbon emissions. The consumption of oil products is closely related to the transportation industry and is one of the main sources of carbon emissions. Therefore, it is also an important data for analyzing carbon emissions. The analysis of natural gas consumption helps to understand the progress of the energy transition and the effects of emissions reduction. Data related to new energy power directly reflects a region's efforts in reducing carbon emissions and promoting energy transition.

In summary, these indicators are of great significance for a comprehensive assessment of regional carbon emissions, identifying potential for reduction, formulating reduction strategies, and monitoring the effects of reduction. By collecting and analyzing these data, policymakers can formulate environmental policies more scientifically, enterprises can implement emission reduction measures more effectively, and the general public can have a clearer understanding of carbon emissions and adopt corresponding low-carbon lifestyles. This section will predict carbon emissions from 2024 to 2060 based on these data to provide an in-depth analysis of the dual carbon issue. Use the collected data to draw a scatter plot as shown below [Figure 11](#).

To examine the relationship between photovoltaic (PV) power generation and carbon emissions more systematically, this study collected regional data on carbon emissions and related socio-economic and energy variables, including GDP, total energy consumption, household consumption, coal consumption, oil consumption, natural gas consumption, and renewable energy power. These indicators jointly reflect economic activity, energy-use structure, and energy transition progress, and therefore provide



an appropriate basis for carbon-emission analysis and long-term projection.

Figure 11 presents the descriptive trends and correlation structure of the key variables. The time-series plots show an overall upward trend in carbon emissions, GDP, and energy consumption over the sample period, while renewable energy power increases more rapidly in the later years, indicating an acceleration of energy transition. The correlation heatmap further shows that carbon emissions are strongly positively correlated with energy consumption (0.9728) and moderately-to-strongly correlated with GDP (0.8720), suggesting that economic expansion and energy demand remain major drivers of emissions. Renewable energy power is positively correlated with GDP (0.9064) and carbon emissions (0.7746), which reflects the fact that renewable-energy expansion has occurred alongside economic growth and rising overall energy demand. These results provide an empirical basis for the subsequent carbon-emission projection analysis for 2024–2060.

## 5.2 Relational model construction

Since within the four indicators, data on GDP and energy consumption for 2023–2060 are known by referring to relevant government documents. Therefore, based on the results of the correlation analysis, a relationship model of new energy power with GDP and energy consumption was constructed. The results are as shown below Table 7:

Based on the analysis of the new energy power production model, this paper yields the following key results: the coefficient of the intercept term is 557.15, but its p-value is 0.52001, indicating that the intercept is statistically insignificant, that is, without considering GDP and energy consumption, the average of the new energy power production predicted by the model is not significantly different from zero. The coefficient of GDP is 0.022264, the standard error is 0.0064502, the t-statistic is 3.4516, and the corresponding p-value is 0.0086744, indicating that the impact of GDP on new energy power is

statistically significant and positively correlated with new energy power output. The coefficient of energy consumption is  $-0.05344$ , the standard error is 0.041763, the t-statistic is  $-1.2796$ , and the corresponding p-value is 0.23654, indicating that the impact of energy consumption on new energy power is not statistically significant, that is, there is no significant linear relationship between the change in energy consumption and the change in new energy power output.

From the perspective of the goodness of fit of the model, the root mean square error (RMSE) is 111, indicating that the difference between the actual value and the model's predicted value is relatively small, and the model's fit is good. The R-square was 0.852, indicating that the model accounted for 85.2% of the total variation. The adjusted R-square was 0.814, which remained high after adjusting for degrees of freedom, suggesting that the model had strong explanatory power. The F-statistic is 22.9, corresponding to a p-value of 0.000486, indicating that at least one independent variable has a significant linear effect on the output of new energy power.

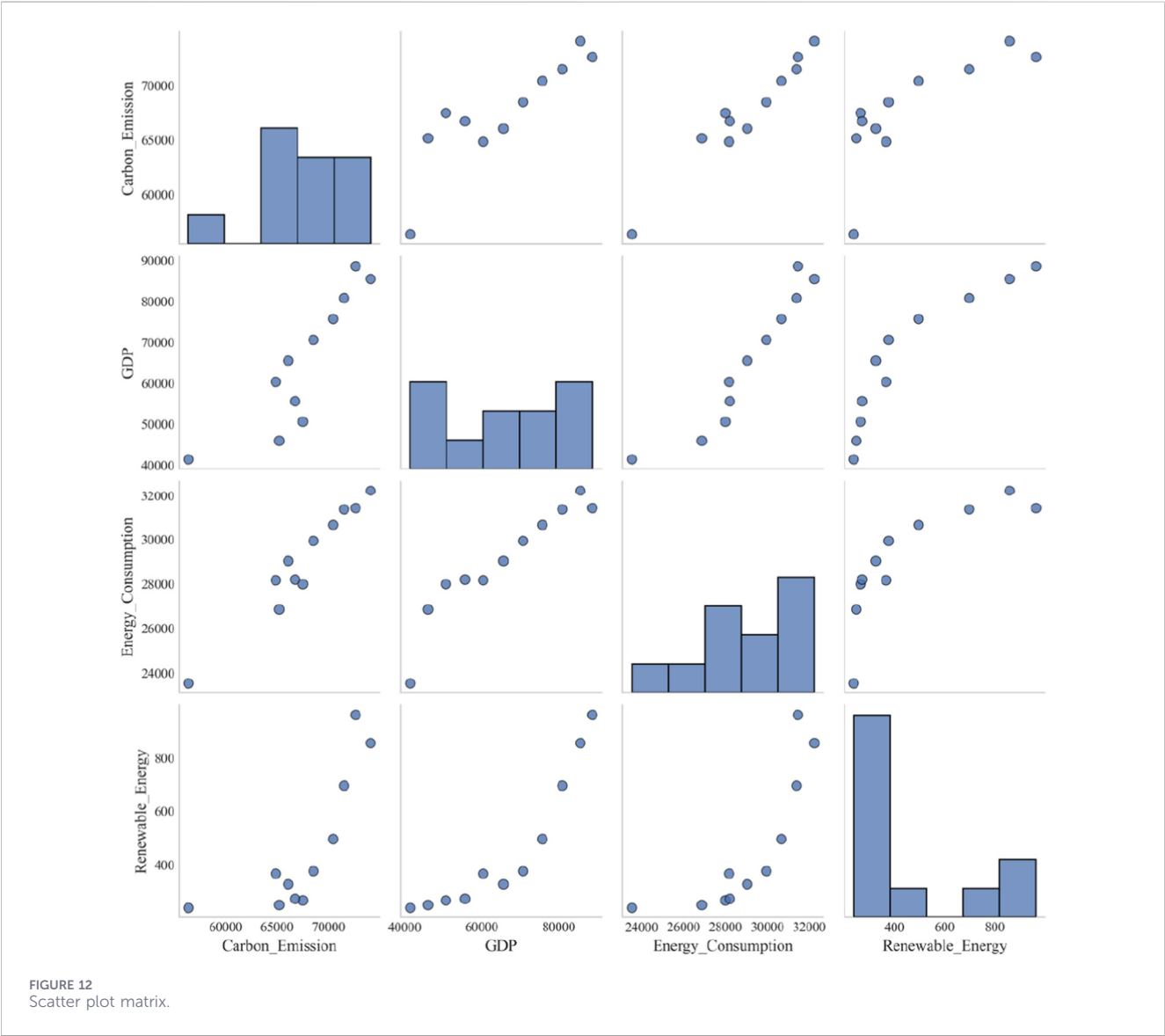
Construct a regression prediction model with carbon emissions as the dependent variable (Y) and other indicators as independent variables (X). Descriptive statistics were first performed on the four indicators, and a scatter plot matrix was plotted as shown below Figure 12.

This is a scatter plot matrix showing the relationship between carbon emissions, GDP, energy consumption and renewable energy generation. It can be seen from the graph that there is a certain positive correlation between carbon emissions and GDP, that is, as GDP increases, carbon emissions also show an upward trend, which may be due to the increase in energy demand during economic development leading to more carbon emissions. At the same time, there is a clear positive correlation between carbon emissions and energy consumption, with an increase in energy consumption directly leading to an increase in carbon emissions. On the other hand, there is a certain negative correlation between renewable energy generation and carbon emissions. As renewable energy generation increases, carbon emissions decrease, indicating that

TABLE 7 Estimation coefficient table.

| Variables   | Estimate | SE        | t-Statistics | p value   |
|-------------|----------|-----------|--------------|-----------|
| (Intercept) | 557.15   | 828.06    | 0.67284      | 0.52001   |
| x1          | 0.022264 | 0.0064502 | 3.4516       | 0.0086744 |
| x2          | −0.05344 | 0.041763  | −1.2796      | 0.23654   |

Number of observations: 11 degrees of error: 8; Root mean square error: 111; R square: 0.852, adjust R square 0.814; F-statistic (constant model): 22.9, p = 0.000486.



the use of renewable energy helps to reduce carbon emissions to some extent. In addition, there is a positive correlation between GDP and energy consumption. Economic development is accompanied by an increase in energy consumption, and there is also a positive correlation between GDP and renewable energy generation, which may imply that economic development promotes the utilization of renewable energy. There is also a positive correlation between energy consumption and renewable energy generation, possibly because as energy consumption increases, so does the demand

for renewable energy. Overall, the relationship between these variables suggests that energy consumption is closely related to carbon emissions during economic development, and the use of renewable energy can relieve the pressure of carbon emissions to some extent and promote sustainable development.

Therefore, a multiple linear regression model was constructed and estimated, and the results are presented below Table 8.

According to the estimated coefficient results in the table, the estimated coefficient for the Intercept term of this model is −4298.5,

TABLE 8 Estimation coefficient table.

| Variables          | Estimate | SE       | t-Statistics | p value    |
|--------------------|----------|----------|--------------|------------|
| (Intercept)        | −4298.5  | 4531.4   | −0.94859     | 0.37442    |
| GDP                | −0.29049 | 0.054178 | −5.3618      | 0.0010507  |
| Energy_Consumption | 3.0059   | 0.24403  | 12.318       | 5.3345e-06 |
| Renewable_Energy   | 7.908    | 1.8822   | 4.2014       | 0.0040286  |

Number of observations: 11, error degrees of freedom: 7; Root mean square error: 593; Party R: 0.989, adjust Party R to 0.985; F-statistic (constant model): 220,  $p = 2.76e-07$ .

but its p-value is 0.37442, greater than the significance level of 0.05, indicating that the intercept term is statistically insignificant. For the independent variable GDP, the estimated coefficient is −0.29049, the standard error (SE) is 0.054178, the t-stat is −5.3618, and the p value is 0.0010507, which is less than 0.05, indicating that GDP has a significant negative impact on the dependent variable. The estimated coefficient for Energy\_Consumption is 3.0059, the standard error is 0.24403, the t-statistic is 12.318, and the p-value is 5.3345e-06, which is much less than 0.05, indicating that energy consumption has a significant positive effect on the dependent variable. The estimated coefficient for Renewable\_Energy is 7.908, the standard error is 1.8822, the t-statistic is 4.2014, and the p-value is 0.0040286, which is less than 0.05, indicating that renewable\_energy also has a significant positive impact on the dependent variable. Overall, an increase in GDP would significantly reduce the value of the dependent variable, while an increase in energy consumption and renewable energy would significantly increase the value of the dependent variable. These results provide an important reference for the formulation of relevant policies.

### 5.3 Projections of carbon emissions without photovoltaic power generation

Establish models of correlations between carbon emissions and population, GDP, and energy consumption. A linear regression model was established based on the processed data and solved by SPSSPRO, and the results are shown in Table 9.

According to the regression analysis results in the table, the model has a strong ability to explain the dependent variable, with an  $R^2$  value of 0.989, indicating that the independent variable can explain the changes in the dependent variable very well. For each independent variable, the non-normalized coefficient B of population is −18.798, the normalized coefficient Beta is −0.778, the t value is −4.04, and the P value is 0.005, which is less than the significance level of 0.05, indicating that population has a significant negative impact on the dependent variable, and its VIF value is 23.384. There is a serious problem of multicollinearity; The non-normalized coefficient B of gross domestic product is −0.017, the normalized coefficient Beta is −0.058, the t value is −0.409, the P value is 0.695, greater than 0.05, indicating that the impact of gross domestic product on the dependent variable is statistically insignificant. Its VIF value is 12.813, and there is also a certain degree of multicollinearity. The non-standardized coefficient B of total energy consumption is 3.453, the standardized coefficient Beta is 1.781, the t value is

10.997, and the P value is 0.000, which is much less than 0.05, indicating that total energy consumption has a significant positive impact on the dependent variable, with a VIF value of 16.536, and there is also a problem of multicollinearity. Overall, total energy consumption has a significant positive effect on the dependent variable, population has a significant negative effect, but gross domestic product has no significant effect.

Since the initial analysis found some collinearity among the indicators. Here, the VIF value represents the severity of multicollinearity and is used to test whether the model is collinear. The VIF values in the table are all large, so ridge regression is used here to establish the association model. To solve the above problem, this paper needs to transform the undetermined problem into a well-determined one by adding a small constant value to the diagonal element of the matrix  $X^T X$  and then taking its inverse to obtain the coefficients  $\lambda$ :

The formula for the ridge regression coefficients  $\hat{\beta}_{\text{ridge}}$  is given by:

$$\hat{\beta}_{\text{ridge}} = (X^T X + \lambda I_n)^{-1} X^T Y$$

where:  $X$  is the matrix of predictor variables (design matrix).  $Y$  is the vector of response variables (target values).  $\lambda$  is A small positive constant, often referred to as the ridge parameter (regularization parameter). It controls the amount of shrinkage applied to the coefficients.  $I_n$  is the identity matrix of size  $n$ , where  $n$  is the number of predictors (or features). The identity matrix has ones along the diagonal and zeros elsewhere.  $(X^T X + \lambda I_n)^{-1}$  is the matrix inversion of the modified design matrix. Adding  $\lambda I_n$  to the matrix  $X^T X$  ensures that the matrix is invertible, even when multicollinearity exists.

This formula effectively adds a shrinkage penalty to the regression model. The higher the value of  $\lambda$ , the greater the penalty applied to the coefficients, which reduces their size and mitigates multicollinearity.

The cost function in ridge regression is composed of two parts:

The residual sum of squares (RSS), which measures the fit of the model. The penalty term, which controls the size of the coefficients to reduce multicollinearity. The cost function is:

$$J_{\beta}(\beta) = \text{RSS} + \lambda \sum_{j=1}^n \beta_j^2 = \text{RSS} + \lambda \|\beta\|^2$$

where: RSS (Residual Sum of Squares) measures how well the model fits the data. Here,  $y_i$  is the observed value for the  $i$ -th sample, and  $X_i \beta$  is the predicted value for the same sample.  $\lambda \|\beta\|^2$  is the penalty

TABLE 9 Results of linear regression.

| Variables                | Non-standardized coefficients |                | Standardized coefficients | t      | P         | VIF    | R <sup>2</sup> |
|--------------------------|-------------------------------|----------------|---------------------------|--------|-----------|--------|----------------|
|                          | B                             | Standard error | Beta                      |        |           |        |                |
| Constant                 | 123996.548                    | 30996.85       | -                         | 4      | 0.005 *** | -      | 0.989          |
| Population               | -18.798                       | 4.653          | -0.778                    | -4.04  | 0.005 *** | 23.384 |                |
| Gross domestic product   | -0.017                        | 0.043          | -0.058                    | -0.409 | 0.695     | 12.813 |                |
| Total energy consumption | 3.453                         | 0.314          | 1.781                     | 10.997 | 0.000 *** | 16.536 |                |

TABLE 10 Results of ridge regression analysis.

| K = 0.045                | Non-standardized coefficients |                | Standardized coefficients | t      | P         | R <sup>2</sup> | Adjust R <sup>2</sup> | F      |
|--------------------------|-------------------------------|----------------|---------------------------|--------|-----------|----------------|-----------------------|--------|
|                          | B                             | Standard error | Beta                      |        |           |                |                       |        |
| Constant                 | 27827.599                     | 26866.457      | -                         | 1.036  | 0.335     | 0.951          | 0.930                 | 45.438 |
| Population               | -1.987                        | 3.83           | -0.082                    | -0.519 | 0.620     |                |                       |        |
| Gross domestic product   | 0.004                         | 0.048          | 0.015                     | 0.093  | 0.929     |                |                       |        |
| Total energy consumption | 1.926                         | 0.311          | 0.994                     | 6.191  | 0.000 *** |                |                       |        |

term that shrinks the coefficients. The term  $\|\beta\|^2$  is the sum of the squares of the coefficients  $\beta_j$ .

Adding this term discourages large values for the coefficients and helps stabilize the estimates, especially when multicollinearity exists among the predictors. In matrix form, the cost function becomes:

$$J_{\beta}(\beta) = \sum_{i=1}^p (y_i - X_i\beta)^2 + \lambda \sum_{j=1}^n \beta_j^2$$

This expression includes:  $(y_i - X_i\beta)^2$ , which represents the squared error for the  $i$ -th data point. The sum  $\sum_{j=1}^n \beta_j^2$ , which is the penalty term that applies to the coefficients. The regularization term  $\lambda \sum_{j=1}^n \beta_j^2$  penalizes large coefficients, which helps in dealing with the multicollinearity problem by shrinking the coefficients toward zero without completely eliminating them.

First, use the ridge trail map to determine the K value. The principle for choosing the K value is the minimum K value when the standardized regression coefficients of each independent variable tend to stabilize. In general, the smaller the K value, the smaller the deviation. We finally, K = 0.045 was determined by the variance expansion factor method, and the ridge regression analysis results are shown in Table 10.

According to the regression analysis results in the table, the model has a strong ability to explain the dependent variable, with an R<sup>2</sup> value of 0.951 and an adjusted R<sup>2</sup> of 0.93, indicating that the independent variable can explain the changes in the dependent variable very well. The F value is 45.438, and the corresponding P value is 0.000, which is less than the significance level of 0.05, indicating that the entire regression model is significant.

For each independent variable, the population's non-normalized coefficient B is -1.987, the normalized coefficient Beta is -0.082, the t value is -0.519, and the P value is 0.62, greater than 0.05, indicating that the population's influence on the dependent variable is statistically

insignificant. The non-standardized coefficient B of gross domestic product was 0.004, the standardized coefficient Beta was 0.015, the t value was 0.093, and the P value was 0.929, all greater than 0.05, indicating that the impact of gross domestic product on the dependent variable was also not significant. The non-standardized coefficient B of total energy consumption is 1.926, the standardized coefficient Beta is 0.994, the t value is 6.191, and the P value is 0.000, which is less than 0.05, indicating that total energy consumption has a significant positive impact on the dependent variable.

$$w_{CO2i} = 27827.599 - 1.987x_{Pi} + 0.004x_{GZi} + 1.926x_{NZi}$$

In the formula above is the model of the correlation between carbon emissions and population, GDP, and energy consumption. At the same time, to explore the relationship between carbon emissions and various energy-consuming sectors (industrial consumption sector, building consumption sector, transportation consumption sector, residential consumption, agricultural and forestry consumption sector) as well as energy supply sectors. Establish a multiple linear regression model of total energy consumption with each energy consumption sector (industrial consumption, building consumption, transportation consumption, residential consumption, agricultural and forestry consumption) and energy supply sector (Yao et al., 2024). The results of the process analysis are presented in the appendix, and here only the final relationship is shown.

$$x_{NZi} = 3916.326 + 8.585x_{NNi} + 0.499x_{NGi} + 0.281x_{NYi} + 0.763x_{Nji} + 2.399x_{NIi} + 0.778x_{NPi}$$

Using this relational formula for prediction, the result of carbon emission prediction without photovoltaic power generation is obtained.



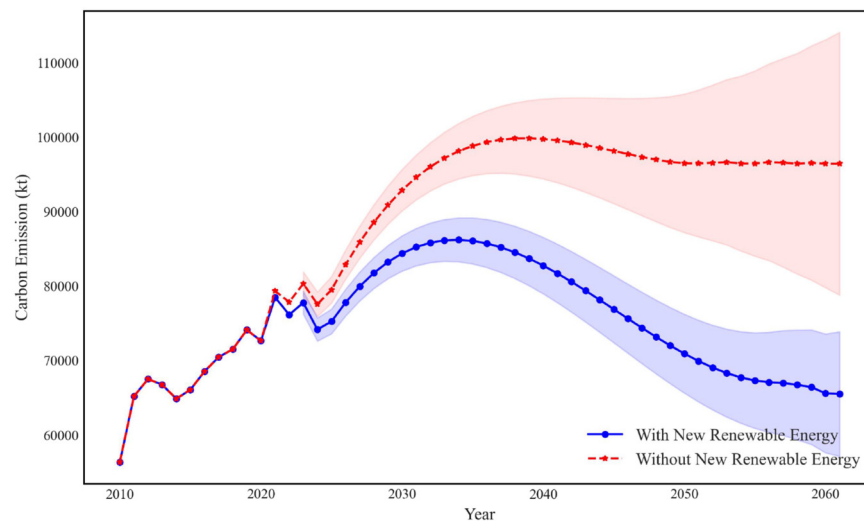


FIGURE 13  
Comparison figure.

## 5.4 Data comparison

Based on Sections 5.3, 5.4, carbon emission data with or without the influence of new energy power data, in order to draw the visualization results more intuitively, this paper draws a line graph for comparison and analysis. As shown below Figure 13.

This chart shows trends in carbon emissions over time under two different scenarios, spanning from 2010 to 2070. There are two curves in the graph: one is a solid blue line representing the change in carbon emissions With New Renewable Energy; The other is a red dotted line, representing the change in carbon emissions Without New Renewable Energy. It can be seen from the graph that without the introduction of new renewable energy (the red dotted line), carbon emissions rose rapidly between 2010 and 2020, then peaked between 2020 and 2030, and then began to decline gradually, but at a slower pace, and stabilized after 2050, remaining at a high level. In the case of the introduction of new renewable energy (blue solid line), carbon emissions also showed an upward trend from 2010 to 2020, but the upward trend slowed significantly after 2020 and began to decline rapidly after reaching a peak around 2030. By around 2050, emissions would have fallen below the scenario without the introduction of renewable energy and continued to decline over the following decades, reaching a lower level by 2070.

## 6 Conclusion

This study systematically analyzed key issues such as power supply forecasting, power station site selection optimization, power generation increase and carbon emission impact of photovoltaic power generation by constructing a multi-dimensional index system and multiple mathematical models. First, based on principal component analysis and particle swarm optimization, a power supply prediction model was constructed to verify the synergy of economic, energy, population and environmental indicators, and to improve data reliability through dimensionality reduction and outlier correction. Secondly, cubic polynomial regression and ridge regression methods were used to

predict the trends of power production and carbon emissions from 2024 to 2060. The results showed that the introduction of photovoltaic power generation could significantly reduce the total carbon emissions, verifying its core role in achieving the “dual carbon” goals. The study innovatively combined t-SNE dimensionality reduction with an improved particle swarm optimization algorithm to optimize the efficiency of solving complex nonlinear problems and provide data-driven decision support for regional energy planning.

However, the model has some limitations, primarily its adaptability to policy changes and market fluctuations. The lack of incorporation of dynamic policy factors and multi-source heterogeneous data in the model may affect its prediction accuracy under rapidly changing real-world conditions. Therefore, future studies should focus on integrating these dynamic elements to enhance the model's robustness and predictive power. Incorporating more diverse data sources, such as policy shifts, market behavior, and local or regional conditions, would further improve the model's real-world applicability.

Overall, this study provides a theoretical basis and technical path for the large-scale application and sustainable development of photovoltaic power generation. The model offers significant reference value for promoting the transformation of energy structures, although improvements are necessary to ensure its full adaptability to future challenges.

## Data availability statement

The original contributions presented in the study are included in the article/Supplementary Material, further inquiries can be directed to the corresponding author.

## Author contributions

PF: Visualization, Funding acquisition, Writing – original draft, Supervision, Validation, Software. ZC: Investigation, Methodology,

Writing – review and editing, Formal Analysis, Software, Data curation, Validation, Project administration. YW: Writing – review and editing, Data curation, Conceptualization, Writing – original draft, Formal Analysis.

## Funding

The author(s) declared that financial support was received for this work and/or its publication. The paper was supported by the National Social Science Fund of China under grant number 22BJY259.

## Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

## Generative AI statement

The author(s) declared that generative AI was not used in the creation of this manuscript.

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